

Störmer's Rule

Mathematical Derivation from First Principles

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1 Problem Statement

We consider a second-order ordinary differential equation of the form

$$y'' = f(x, y), \quad y(x_0) = y_0, \quad y'(x_0) = v_0, \quad (1)$$

where $y : \mathbb{R} \rightarrow \mathbb{R}^n$, $f : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, and crucially f does not depend on y' .

Remark 1 (Physical motivation). *Equation (1) describes any purely position-dependent force: gravitational n -body, Keplerian orbits, conservative mechanical systems, and the Störmer problem (charged particle in a magnetic field) that originally motivated the method. For the two-body orbital problem: $f(\mathbf{r}) = -\mu \mathbf{r}/|\mathbf{r}|^3$.*

Goal. Advance the solution from x_0 to $x_0 + H$ by subdividing H into M equal sub-steps of size $h = H/M$, in a way that:

1. exploits the absence of y' in f ,
2. produces an *even* error series in h (enabling Bulirsch-Stoer Richardson extrapolation), and
3. requires only $M/2$ new evaluations of f per column (half the cost of applying the standard modified midpoint rule to the equivalent first-order system).

2 The Störmer Three-Point Recurrence

2.1 Derivation from Taylor's Theorem

Write the Taylor expansions of y centered at $x_m = x_0 + mh$:

$$y(x_m + h) = y_m + hy'_m + \frac{h^2}{2}y''_m + \frac{h^3}{6}y'''_m + \frac{h^4}{24}y^{(4)}_m + \mathcal{O}(h^5), \quad (2)$$

$$y(x_m - h) = y_m - hy'_m + \frac{h^2}{2}y''_m - \frac{h^3}{6}y'''_m + \frac{h^4}{24}y^{(4)}_m + \mathcal{O}(h^5). \quad (3)$$

Adding (2) and (3) cancels all odd powers:

$$y(x_m + h) + y(x_m - h) = 2y_m + h^2y''_m + \frac{h^4}{12}y^{(4)}_m + \mathcal{O}(h^6). \quad (4)$$

Substituting $y''_m = f(x_m, y_m)$ and rearranging:

$$\boxed{y_{m+1} = 2y_m - y_{m-1} + h^2f(x_m, y_m) + \mathcal{O}(h^4).} \quad (5)$$

This is the **Störmer three-point formula**: given two successive solution values y_{m-1} and y_m , the next value y_{m+1} requires a single evaluation of f and is accurate to $\mathcal{O}(h^4)$ locally.

Remark 2 (No velocity in the recurrence). *Equation (5) uses y_{m-1} , y_m , and $f(x_m, y_m)$ only. The velocity y' has been eliminated algebraically. The price is the need for two starting values.*

2.2 The δ -Increment Formulation

Define the position increment $\delta_m := \theta_{m+1} - \theta_m$, the change in the numerical solution over one sub-step. From (5):

$$\begin{aligned} \delta_m &= y_{m+1} - y_m \\ &= (2y_m - y_{m-1} + h^2f_m) - y_m \\ &= (y_m - y_{m-1}) + h^2f_m \\ &= \delta_{m-1} + h^2f_m. \end{aligned} \quad (6)$$

This telescoping relation propagates δ forward with only h^2f_m added at each step. The velocity information is encoded in the initial δ_0 .

Starting value from Taylor: The value $y_1 = y(x_0 + h)$ from (2) with $m = 0$:

$$y_1 = y_0 + h v_0 + \frac{h^2}{2} f_0 + \mathcal{O}(h^3), \quad (7)$$

so

$$\boxed{\delta_0 = h v_0 + \frac{h^2}{2} f_0.} \quad (8)$$

3 Order Analysis: What “Order p ” Means

The three-point recurrence (5) was derived from Taylor’s theorem. We now ask precisely: *how fast does the global error shrink as $h \rightarrow 0$?* The answer, “order p ”, refers entirely to powers of h — it is **not** the degree of a polynomial fitted to f .

3.1 The HNW Framework

Following Hairer, Nørsett & Wanner [1] (henceforth **HNW**), write the general second-order multistep method for $y'' = f(x, y)$ as

$$\sum_{i=0}^k \alpha_i y_{n+i} = h^2 \sum_{i=0}^k \beta_i f(x_{n+i}, y_{n+i}), \quad \text{HNW (III.10, Eq. 10.19)} \quad (9)$$

with scalar sequences $\{\alpha_i\}$ and $\{\beta_i\}$, $\alpha_k \neq 0$. Define the two **characteristic polynomials**

$$\varrho(\zeta) = \sum_{i=0}^k \alpha_i \zeta^i, \quad \sigma(\zeta) = \sum_{i=0}^k \beta_i \zeta^i. \quad (10)$$

These encode the entire method: ϱ governs stability and σ governs accuracy.

The residual when the *exact* solution is substituted into the method is measured by the **linear difference operator** (HNW, Eq. 10.23, p. 467):

$$\mathcal{L}[y, x, h] = \varrho(E) y(x) - h^2 \sigma(E) y''(x) = \sum_{i=0}^k \left[\alpha_i y(x + ih) - h^2 \beta_i y''(x + ih) \right], \quad (11)$$

where E is the shift operator $E y(x) = y(x + h)$. Along the exact solution $y''(x + ih) = f(x + ih, y(x + ih))$, so $\mathcal{L}$ is a pure function of the smooth exact solution y and the step h .

Theorem 1 (HNW Definition 10.2 and Theorem 10.3, p. 467–468). *Method (9) is **consistent of order p** if*

$$\mathcal{L}[y, x, h] = \mathcal{O}(h^{p+2}) \quad \text{for every sufficiently smooth } y. \quad (12)$$

Three equivalent conditions (HNW Theorem 10.3, p. 468) are:

(i) Moment form. $\sum_i \alpha_i = 0$, $\sum_i i \alpha_i = 0$, and for $q = 2, \dots, p+1$:

$$\sum_{i=0}^k \alpha_i i^q = q(q-1) \sum_{i=0}^k \beta_i i^{q-2}. \quad (13)$$

(ii) Generating-polynomial form.

$$\varrho(e^h) - h^2 \sigma(e^h) = \mathcal{O}(h^{p+2}) \quad \text{as } h \rightarrow 0. \quad (14)$$

(iii) Root form near $\zeta = 1$.

$$\frac{\varrho(\zeta)}{(\log \zeta)^2} - \sigma(\zeta) = \mathcal{O}((\zeta - 1)^p) \quad \text{as } \zeta \rightarrow 1. \quad (15)$$

Why h^{p+2} , not h^p ? The method equates the h^2 -scaled forcing $h^2 f$ to the finite-difference $\sum \alpha_i y_{n+i}$, so the entire equation is naturally $\mathcal{O}(h^2)$. The residual $\mathcal{L}$ is therefore already premultiplied by h^2 ; order p means the *extra* smallness beyond h^2 is h^p more — i.e., h^{p+2} total. The global error, obtained by accumulating $\mathcal{O}(h^{p+2})$ local residuals over $\mathcal{O}(1/h)$ steps and dividing once by the implicit h^2 factor, is $\mathcal{O}(h^p)$ — matching Definition 10.2. This is confirmed by:

Theorem 2 (HNW Theorem 10.6, p. 468 — Global Convergence). *If method (9) is stable and of order p , and the starting values satisfy $y(x_j) - y_j = \mathcal{O}(h^{p+1})$ for $j = 0, \dots, k-1$, then*

$$|y(x_n) - y_n| \leq C h^p \quad \text{for all } 0 \leq hn \leq \text{const}. \quad (16)$$

3.2 Verification for the $k = 2$ Störmer Method

For the three-point formula (5):

$$\alpha_0 = 1, \alpha_1 = -2, \alpha_2 = 1; \quad \beta_0 = 0, \beta_1 = 1, \beta_2 = 0. \quad (17)$$

Hence $\varrho(\zeta) = \zeta^2 - 2\zeta + 1 = (\zeta - 1)^2$ and $\sigma(\zeta) = 1$ (the center-point quadrature rule for f).

Apply the generating-polynomial test (14). Expanding $\varrho(e^h)$ term by term:

$$\varrho(e^h) = e^{2h} - 2e^h + 1 = h^2 + \frac{h^4}{12} + \frac{h^6}{360} + \dots \quad (18)$$

(all odd powers cancel because $e^{2h} - 2e^h + 1$ is symmetric under $h \mapsto -h$ after recentering). With $\sigma(\zeta) = 1$:

$$\varrho(e^h) - h^2 \sigma(e^h) = \underbrace{h^2}_{\text{cancel}} + \frac{h^4}{12} + \frac{h^6}{360} + \dots - h^2 = \frac{h^4}{12} + \frac{h^6}{360} + \dots = \mathcal{O}(h^4). \quad (19)$$

Comparing with condition (12): $h^{p+2} = h^4 \Rightarrow \boxed{p = 2}$. The method is **globally second-order accurate** in h .

The leading residual coefficient $\frac{1}{12}$ is the **error constant** $C = \sigma_2 = \frac{1}{12}$ (HNW Exercise 5, §III.10). It governs the asymptotic error via (HNW Eq. 10.36–10.37, p. 471):

$$\kappa''(x) = \frac{\partial f}{\partial y}(x, y(x)) \kappa(x) - \frac{1}{12} y^{(4)}(x), \quad (20)$$

where $\kappa(x)$ satisfies $y_n - y(x_n) = \kappa(x_n) h^2 + \mathcal{O}(h^3)$. For orbital mechanics, $y^{(4)}$ is the fourth derivative of the position trajectory; a smooth orbit (low eccentricity) keeps $y^{(4)}$ small, making the error constant favorable.

3.3 The Rounding-Error Trap: Why the δ -Form Is Mandatory

The polynomial $\varrho(\zeta) = (\zeta - 1)^2$ has a **double root at** $\zeta = 1$. HNW Definition 10.1 (p. 467) allows roots on the unit circle with multiplicity at most 2 — our method sits exactly on this boundary.

What this means in practice (HNW §III.10, pp. 472–473, Fig. 10.3): a *direct* implementation that stores $y_0, y_1, \dots$ and applies the recurrence $y_{n+1} = 2y_n - y_{n-1} + h^2 f_n$ accumulates rounding errors that grow without bound as $h \rightarrow 0$. The double root causes the companion matrix of the difference equation to be no longer power-bounded.

The **fix** is the δ -increment form (HNW Eq. 10.38', p. 472):

$$\delta_n - \delta_{n-1} = h^2 f_n, \quad y_{n+1} - y_n = \delta_n. \quad (21)$$

This splits $\varrho(\zeta)$ as $(\zeta - 1)(\zeta - 1)$ and introduces $\delta_n = y_{n+1} - y_n$ as a new variable. The resulting companion matrix is power-bounded (HNW Eq. 10.30), and rounding errors remain bounded. This is exactly the formulation used in `StormerStep::step()`.

A second pitfall (HNW p. 473, note b): when computing *starting values* $\delta_0, \delta_1, \dots$, do **not** form the differences $(y_1 - y_0)/h$, $(y_2 - y_1)/h$ from separately computed RK values — that difference is numerically unstable. Instead, accumulate the increments directly from the sub-steps of the RK starting procedure, just as `StormerStep` does internally.

4 Störmer's Modified Midpoint Algorithm

Notation. We introduce θ_m to denote the **numerical approximation** to the true solution $y(x_m)$ at sub-step m , where $x_m = x_0 + mh$. The exact solution is $y(x_m)$; its approximation is θ_m . The algorithm computes a sequence of approximations:

$$\theta_0 \approx y(x_0), \quad \theta_1 \approx y(x_1), \quad \dots, \quad \theta_M \approx y(x_0 + H).$$

The velocity approximation at the final point is denoted $z_M \approx y'(x_0 + H)$.

Why θ instead of y ? In numerical analysis, it is standard to distinguish the *exact* solution $y(x)$ (unknown, continuous) from the *computed* approximation θ_m (known, discrete). The error at step m is $y(x_m) - \theta_m$. This distinction becomes crucial when analyzing convergence and error propagation.

The complete algorithm for M sub-steps of size $h = H/M$ is:

1. Initialize.

$$\theta_0 = y_0, \quad (22)$$

$$\delta_0 = h v_0 + \frac{h^2}{2} f(x_0, \theta_0). \quad (23)$$

Set $\theta_1 = \theta_0 + \delta_0$.

2. Sub-step loop for $m = 1, \dots, M - 1$:

$$f_m = f(x_0 + mh, \theta_m), \quad (24)$$

$$\delta_m = \delta_{m-1} + h^2 f_m, \quad (25)$$

$$\theta_{m+1} = \theta_m + \delta_m. \quad (26)$$

3. **Final derivative** (velocity estimate at $x_0 + H$):

$$z_M = \frac{\delta_{M-1}}{h} + \frac{h}{2} f(x_0 + H, \theta_M). \quad (27)$$

The output is $(\theta_M, z_M) \approx (y(x_0 + H), y'(x_0 + H))$, the numerical approximations to the true position and velocity at $x_0 + H$, with local truncation error $\mathcal{O}(h^2)$.

Total function evaluations of f : one per sub-step $m = 0, \dots, M$, i.e. $M + 1$ total, but f_0 is supplied from the previous step, so M new evaluations.

5 Equivalence to Störmer-Verlet / Leapfrog

Define the half-integer velocity $v_{m+1/2} := \delta_m/h$. Rewriting the algorithm in terms of v :

$$v_{1/2} = \frac{\delta_0}{h} = v_0 + \frac{h}{2} f_0 \quad (\text{half-kick}), \quad (28)$$

$$\theta_{m+1} = \theta_m + h v_{m+1/2} \quad (\text{drift}), \quad (29)$$

$$v_{m+3/2} = v_{m+1/2} + h f_{m+1} \quad (\text{full kick, } m = 0, \dots, M-2), \quad (30)$$

and the final velocity:

$$z_M = v_{M-1/2} + \frac{h}{2} f_M. \quad (31)$$

Equation (31) can be read as: take the half-integer velocity $v_{M-1/2}$, and kick it forward by a half step to return to integer time M . Equivalently:

$$z_M = \frac{v_{M-1/2} + v_{M+1/2}}{2}, \quad \text{where } v_{M+1/2} = v_{M-1/2} + h f_M. \quad (32)$$

So z_M is the arithmetic mean of the bracketing half-integer velocities, a second-order approximation to $y'(x_0 + H)$.

Remark 3 (Symplecticity). *The drift-kick-drift structure (28)–(30) is the Störmer-Verlet integrator, which is symplectic for Hamiltonian systems $H = \frac{1}{2}|v|^2 + V(y)$ with $f = -\nabla V$. It preserves a modified Hamiltonian exactly to all orders. Importantly, it does NOT suffer from secular energy drift.*

6 Even Error Series: Gragg's Theorem

Theorem 3 (Gragg 1965). *Let $y(x)$ be sufficiently smooth. For Störmer's modified mid-point rule applied to $y'' = f(x, y)$ with M sub-steps of $h = H/M$ and the starting/ending conventions (8)–(27), the global errors admit expansions in even powers of h only:*

$$y(x_0 + H) - \theta_M = a_2(H) h^2 + a_4(H) h^4 + a_6(H) h^6 + \dots, \quad (33)$$

$$y'(x_0 + H) - z_M = b_2(H) h^2 + b_4(H) h^4 + b_6(H) h^6 + \dots, \quad (34)$$

where the coefficients $a_{2k}(H)$, $b_{2k}(H)$ are independent of M .

Sketch of proof (time-reversibility argument). The Störmer recurrence (22)–(27) is *time-reversible*: reversing $h \rightarrow -h$ and the sequence of θ_m recovers the original sequence. A time-reversible method cannot accumulate odd powers of h in its global error, because odd-power terms change sign under $h \rightarrow -h$ while the exact solution does not. More formally, one expands $\theta_m(h)$ as a power series in h and shows that all odd-power terms cancel identically (see Hairer, Nørsett & Wanner, Vol. I, §II.8 for the complete argument).

Corollary 1 (Richardson extrapolation efficiency). *Because the error series (33) contains only $h^2, h^4, \dots$, a single Richardson extrapolation step using results at h and $h/2$ eliminates the h^2 term and yields $\mathcal{O}(h^4)$ accuracy. In general, k extrapolation steps yield $\mathcal{O}(h^{2k+2})$ accuracy from $\mathcal{O}(h^2)$ base integrations. This is the foundation of the Bulirsch-Stoer method.*

7 Richardson Extrapolation Setup (Bulirsch-Stoer)

Evaluate the modified midpoint rule with sub-step counts $M_1 < M_2 < M_3 < \dots$ (the Bulirsch-Stoer sequence: $\{2, 4, 6, 8, 12, 16, \dots\}$), giving approximations $T_{j,0} := \theta_{M_j}(H)$.

Define the extrapolation table recursively:

$$T_{j,k} = T_{j,k-1} + \frac{T_{j,k-1} - T_{j-1,k-1}}{(M_j/M_{j-k})^2 - 1}, \quad k = 1, 2, \dots, j. \quad (35)$$

Each extrapolation step eliminates the next order in h^2 . The diagonal $T_{k,k}$ converges to $y(x_0 + H)$ as k increases.

Remark 4 (Computational cost vs. accuracy). *For the first-order system $[y', f(x, y)]^T$ of dimension $2n$, the modified midpoint rule needs M evaluations of f per column. Störmer's rule needs only $M/2$ evaluations per column (the NR3 code uses $n_{\text{eff}} = M/2$ leapfrog steps of $h_{\text{eff}} = 2h = 2H/M$, as shown below) for the same total interval H . Since f can be expensive (e.g., gravitational force with many bodies), this factor of two is significant.*

8 Analysis of the NR3 Implementation

8.1 Parameterization

NR3's `StepperStoerm::dy()` uses:

- `nstep` = M (from Bulirsch-Stoer sequence $\{2, 4, 6, 8, \dots\}$).
- `h` = H/M (unit sub-step).
- `h2` = $2h = 2H/M$ (actual loop step, double the unit sub-step).
- `nstp2` = $M/2$ (number of loop iterations).

8.2 Algorithm in NR3 variables

Initialization:

$$\begin{aligned} \mathbf{ytemp}[i] &\leftarrow y[i] & \theta_0 &= y_0, \\ \mathbf{ytemp}[n+i] &\leftarrow y'[i] + h \cdot f_0 & v^* &= v_0 + h f_0 = v_{1/2} \cdot \underbrace{2}_{h_{\text{eff}}/h}. \end{aligned}$$

In other words, the stored “velocity” $\mathbf{ytemp}[n+i]$ is the half-integer velocity $v_{1/2} = v_0 + (h/2) f_0$ scaled by 2: $\mathbf{ytemp}[n+i] = 2 v_{1/2}$. Then the drift step below with factor $h_{\text{eff}} = 2h$ is:

$$\theta_1 = \theta_0 + h_{\text{eff}} \cdot \mathbf{ytemp}[n+i]/2 = \theta_0 + h_{\text{eff}} v_{1/2} = \theta_0 + 2h \left(v_0 + \frac{h}{2} f_0 \right) = \theta_0 + 2h v_0 + h^2 f_0.$$

Alternatively, viewing the NR3 scheme directly as Störmer-Verlet with step $h_{\text{eff}} = 2h$:

$$\text{Loop } m = 1, \dots, n_{\text{eff}} : \quad \begin{cases} \theta_m = \theta_{m-1} + h_{\text{eff}} v_{m-1/2}, \\ v_{m+1/2} = v_{m-1/2} + h_{\text{eff}} f(x_0 + m h_{\text{eff}}, \theta_m), \end{cases} \quad m < n_{\text{eff}}. \quad (36)$$

Final velocity: $z = v_{n_{\text{eff}}-1/2} + (h_{\text{eff}}/2) f(x_0 + H, \theta_{n_{\text{eff}}})$.

8.3 Cost comparison

Method	State dimension	Evals of f per column	Error order
Midpoint (first-order 2n-system)	$2n$	M	$\mathcal{O}(h^2)$
Störmer (NR3)	n	$M/2$	$\mathcal{O}(h_{\text{eff}}^2)$

Both methods embed in a Bulirsch-Stoer extrapolation with an even-power error series and converge to the same accuracy. Störmer uses half the evaluations of f at the cost of doubling the effective sub-step size, which the extrapolation corrects.

9 Is Inheritance Necessary?

NR3’s design uses `struct StepperStoerm : StepperBS<D> solely` for code reuse of the Bulirsch-Stoer extrapolation machinery (`odeint` driver, column sequence, Richardson table, stepsize control). The method `dy()` is entirely independent of the base class algorithms; it overrides only the *sub-step integration* while reusing everything else.

From first principles, the minimal reusable components are:

1. **StormerStep**: executes n_{eff} leapfrog sub-steps over a given interval H . No base class required. Template on `DerivativeType` and `ContainerT`.
2. **Error estimation**: step-doubling (compare n_{eff} steps vs. $2 n_{\text{eff}}$ steps) or Richardson extrapolation as a separate, composable component.
3. **Step-size control**: compose with the existing `ComputePIStepSize / PIStepSizeControl` classes already present in the library.

Composition over inheritance, exactly consistent with how `CalculateNewYAndError` and `IntegrateWithPIControl` were designed.

10 Harmonic Oscillator Test Case

The standard test for Störmer’s rule is the harmonic oscillator:

$$y'' = -\omega^2 y, \quad y(0) = 1, \quad y'(0) = 0, \quad \omega = 1. \quad (37)$$

Exact solution: $y(t) = \cos t$, $y'(t) = -\sin t$, with conserved energy:

$$E(t) = \frac{1}{2}(y'(t))^2 + \frac{1}{2}(y(t))^2 = \frac{1}{2}. \quad (38)$$

Störmer-Verlet (symplectic) conserves a modified energy exactly, so the energy error remains bounded for all time (no secular drift), unlike non-symplectic methods.

Expected convergence. For a single step of size $H = 0.5$ with n_{eff} leapfrog sub-steps of $h_{\text{eff}} = H/n_{\text{eff}}$:

$$|y(H) - \theta_M| \approx \frac{H^3}{24} |y'''(0)| \cdot h_{\text{eff}}^2 = \frac{H^3 \omega^3}{24} \cdot h_{\text{eff}}^2 = \mathcal{O}(h_{\text{eff}}^2).$$

Doubling n_{eff} (halving h_{eff}) should reduce the error by a factor of 4. This is the convergence test in the unit tests.

11 Numerov’s Method: The Implicit Upgrade

Section 3 showed that the three-point stencil (5) achieves order 2 because $\varrho(e^h) - h^2 \sigma(e^h) = \mathcal{O}(h^4)$ with $\sigma(\zeta) = 1$. The same stencil on the left-hand side, but a richer choice of σ , can reach $\mathcal{O}(h^6)$ — i.e. order 4 — with *no extra history*.

11.1 Notation Bridge: HNW Polynomials vs NR3/Our Variables

The analysis in §3 introduced Hairer–Nørsett–Wanner (HNW) notation. Here is a translation table bridging their polynomial formalism with the discrete variables used earlier in this document and in NR3:

Symbol	Meaning	Where it appears
$\varrho(\zeta)$	First characteristic polynomial $\sum_{i=0}^k \alpha_i \zeta^i$	HNW Eq. 10.20, p. 467 (stability)
$\sigma(\zeta)$	Second characteristic polynomial $\sum_{i=0}^k \beta_i \zeta^i$	HNW Eq. 10.20, p. 467 (accuracy)
ζ	Formal variable; $E = \zeta$ is the shift operator: $Ey_n = y_{n+1}$	HNW §III.1
α_i	Coefficients of LHS stencil (position weights)	Eq. (9)
β_i	Coefficients of RHS quadrature (force weights)	Eq. (9)
k	Step number (history length k) Method uses $y_n, \dots, y_{n+k}$	HNW §III.10
h	Step size	Same throughout
H	Total interval (macro-step)	§4
y_n	Exact solution at $x_n = x_0 + nh$	Continuous $y(x_n)$
θ_n	Computed approximation to y_n	§4
f_n	Force $f(x_n, y_n)$ or $f(x_n, \theta_n)$	§4
δ_n	Position increment $\theta_{n+1} - \theta_n$	Eq. (6)
$v_{1/2}$	Half-integer velocity (leapfrog)	Eq. (28)

Connection to our earlier notation. In §2–4 we wrote the method as

$$\theta_{m+1} - 2\theta_m + \theta_{m-1} = h^2 f_m.$$

In HNW polynomial form this is

$$\varrho(\zeta) = \zeta^2 - 2\zeta + 1 = (\zeta - 1)^2, \quad \sigma(\zeta) = 1.$$

The coefficients are $(\alpha_0, \alpha_1, \alpha_2) = (1, -2, 1)$ and $(\beta_0, \beta_1, \beta_2) = (0, 1, 0)$. The shift operator E acts as $E\theta_m = \theta_{m+1}$, so the compact operator form is $\varrho(E)\theta_m = h^2\sigma(E)f_m$.

NR3 notation. NR3 uses ‘nstep’ for the number of sub-steps M , ‘h’ for H/M , ‘h2’ for $2h$, and ‘nstp2’ for $M/2$. Their ‘ytemp[n+i]’ stores the half-integer velocity $v_{m+1/2}$ scaled by 2 (see §??).

11.2 Derivation via the Order Condition

Keep $\varrho(\zeta) = \zeta^2 - 2\zeta + 1$ and seek $\sigma(\zeta) = \beta_0 + \beta_1\zeta + \beta_2\zeta^2$ (symmetric: $\beta_0 = \beta_2$) such that $\varrho(e^h) - h^2\sigma(e^h) = \mathcal{O}(h^6)$.

From (19), $\varrho(e^h) = h^2 + \frac{h^4}{12} + \frac{h^6}{360} + \dots$. We need $h^2\sigma(e^h)$ to match through the h^5 term. Expanding with $\beta_0 = \beta_2 \equiv \beta$, $\beta_1 \equiv \gamma$:

$$h^2\sigma(e^h) = h^2(\beta e^{2h} + \gamma e^h + \beta) = h^2[(2\beta + \gamma) + 2h\beta + h^2(2\beta + \frac{\gamma}{2}) + \dots].$$

Matching through h^5 against $\varrho(e^h)$ forces $2\beta + \gamma = 1$ (the h^2 term), and solving the system for the h^4 match gives $\beta = \frac{1}{12}$, $\gamma = \frac{10}{12}$. Hence:

$$\sigma(\zeta) = \frac{\zeta^2 + 10\zeta + 1}{12}. \quad (39)$$

The resulting method is (HNW Eq. 10.12, p. 464; B. Numerov 1924):

$$\boxed{y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}(f_{n+1} + 10f_n + f_{n-1})}, \quad (40)$$

called **Numerov's method**. The only difference from (5) is the f_{n+1} term on the right: one additional force evaluation at the new position.

11.3 Order Verification

Apply the generating-polynomial test (14):

$$\begin{aligned} h^2\sigma(e^h) &= \frac{h^2}{12}(e^{2h} + 10e^h + 1) \\ &= h^2 + \frac{h^4}{12} + \frac{h^6}{144} + \dots \end{aligned} \quad (41)$$

and recall $\varrho(e^h) = h^2 + \frac{h^4}{12} + \frac{h^6}{360} + \dots$. Subtracting:

$$\varrho(e^h) - h^2\sigma(e^h) = \left(\frac{1}{360} - \frac{1}{144}\right)h^6 + \mathcal{O}(h^8) = -\frac{1}{240}h^6 + \mathcal{O}(h^8) = \mathcal{O}(h^6). \quad (42)$$

Comparing with (12): $h^{p+2} = h^6 \Rightarrow \boxed{p = 4}$. **Numerov's method is globally fourth-order accurate.**

The error constant is $C = \sigma_3^* = 0$ (Table 10.2, HNW p. 464) — the leading error is $-h^6/240 \cdot y^{(6)}$ per step, contributing:

$$\kappa''(x) = \frac{\partial f}{\partial y} \kappa - \frac{1}{240} y^{(6)}(x) \quad (\text{global error: } y_n - y(x_n) = \kappa(x_n) h^4 + \mathcal{O}(h^5)). \quad (43)$$

11.4 The Order-Barrier Context

HNW Theorem 10.4 (p. 468) states that for a stable k -step Störmer method:

$$p \leq k + 2 \text{ if } k \text{ is even,} \quad p \leq k + 1 \text{ if } k \text{ is odd.} \quad (44)$$

For $k = 2$ (our stencil): $p_{\max} = 4$. Numerov saturates this bound — it is the *highest-order stable method* on the three-point stencil. The explicit method (order 2) leaves two orders on the table.

11.5 PECECE Implementation

Numerov’s method is implicit: y_{n+1} appears on both sides via $f_{n+1} = f(x_{n+1}, y_{n+1})$. For nonlinear f (e.g. gravitational force), we solve using **PECECE** (Predict–Evaluate–Correct–Evaluate–Correct–Evaluate):

1. **Predict.** Use the explicit Störmer formula as first guess:

$$y_{n+1}^{(P)} = 2y_n - y_{n-1} + h^2 f_n. \quad (45)$$

2. **Evaluate.** $f_{n+1}^{(P)} = f(x_{n+1}, y_{n+1}^{(P)})$.

3. **Correct.**

$$y_{n+1}^{(1)} = 2y_n - y_{n-1} + \frac{h^2}{12} (f_{n+1}^{(P)} + 10f_n + f_{n-1}). \quad (46)$$

4. **Evaluate.** $f_{n+1}^{(1)} = f(x_{n+1}, y_{n+1}^{(1)})$.

5. **Correct again.**

$$y_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{12} (f_{n+1}^{(1)} + 10f_n + f_{n-1}). \quad (47)$$

6. **Evaluate.** $f_{n+1} = f(x_{n+1}, y_{n+1})$ (stored for next step).

Velocity. Numerov delivers positions. Velocity at integer time is recovered via the leapfrog formula (order 2 in h):

$$v_{n+1} = \frac{y_{n+1} - y_n}{h} + \frac{h}{2} f_{n+1}. \quad (48)$$

Position accuracy is $\mathcal{O}(h^4)$; velocity accuracy is $\mathcal{O}(h^2)$. For applications where velocity precision matters as much as position (e.g. orbit determination), the velocity can be improved via central differences: $v_n \approx (y_{n+1} - y_{n-1})/(2h) + \mathcal{O}(h^2)$ — same order.

Order of PECECE. With an order-2 predictor and order-4 corrector, PECE alone is order 3: the predictor error $\mathcal{O}(h^2)$ contaminates $f_{n+1}^{(P)}$ by $\mathcal{O}(h^2)$, adding $\frac{h^2}{12} \cdot \mathcal{O}(h^2) = \mathcal{O}(h^4)$ per step to the corrector residual, giving $\mathcal{O}(h^3)$ global error. The second correction (PECECE) reduces the contamination to $\mathcal{O}(h^6)$ per step — below the corrector’s own leading error of $\mathcal{O}(h^6)$ — so the full order-4 accuracy is recovered.

11.6 Cost Comparison

Method	Order p	f evals/step	Error constant C
Explicit Störmer ($k = 2$)	2	1	$\sigma_2 = \frac{1}{12}$
Numerov PECECE ($k = 2$)	4	3	$-\frac{1}{240}$

Halving h reduces error by $2^2 = 4$ for explicit Störmer vs. $2^4 = 16$ for Numerov — a $4\times$ gain per step halving. The cost of 3 f -evals vs. 1 is recovered after the first tolerance tightening: to reach the same accuracy, Numerov needs $\approx 2.5\times$ fewer steps than explicit Störmer (since $4^{1/4}/2^{1/2} \approx 1.41$ ratio in step size for equal error).

11.7 Starting Procedure

Numerov is a two-step method: it needs y_0 and y_1 to start. Given initial conditions (y_0, v_0) , one accurate starting step is required. Theorem 2 demands $y(x_1) - y_1 = \mathcal{O}(h^5)$ for the method to retain its full order-4 convergence. Options:

- **Taylor expansion:** $y_1 = y_0 + hv_0 + \frac{h^2}{2}f_0 + \frac{h^3}{6}f'_0 + \frac{h^4}{24}f''_0$ (requires derivative information, gives $\mathcal{O}(h^5)$, satisfies requirement).
- **Runge-Kutta-Nyström order 4:** one step of a 4th-order Nyström method (HNW Table 14.2, p. 285) gives $\mathcal{O}(h^5)$ accuracy, at the cost of 3–4 f -evaluations for the single startup step.
- **Two steps of explicit Störmer** (order 2): gives only $\mathcal{O}(h^3)$, limiting global convergence to $\mathcal{O}(h^2)$. Sufficient only for testing.

`NumerovStep::startup()` uses one step of the leapfrog (Störmer-Verlet) with n_{eff} sub-steps as a high-accuracy starter: for large n_{eff} , this approximates the exact solution to whatever precision is needed. In production, supply a 4th-order starter.

12 Astrodynamics Context: Cowell’s Method, Encke’s Method, and Perturbations

In orbital mechanics, the equations of motion for a satellite under perturbations are:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_p \quad (49)$$

where $\mathbf{a}_p$ is the sum of all perturbing accelerations. Two main approaches exist for numerical integration:

12.1 Cowell’s Method (Direct Integration)

Cowell’s method [3, 4, 5] directly integrates the total acceleration:

$$\dot{\mathbf{r}} = \mathbf{v} \quad (50)$$

$$\dot{\mathbf{v}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_p \quad (51)$$

Advantages:

- Simplicity of formulation and implementation
- Any number of perturbations can be handled simultaneously
- Obtains $\mathbf{r}$ and $\mathbf{v}$ directly

Disadvantages:

- Near large attracting bodies, smaller steps are required (affects time and roundoff error)
- Approximately $10\times$ slower than Encke’s method [3]
- Roundoff error accumulates rapidly in double-precision for interplanetary flights with small step sizes

12.2 Encke’s Method (Osculating Orbit)

Encke’s method [3, 5] integrates the *deviation* from a reference (osculating) two-body orbit rather than the total state. Let $\rho(t)$ be the osculating orbit position (the path the satellite would follow if perturbations vanished at epoch t_0). The deviation $\delta\mathbf{r} = \mathbf{r} - \rho$ satisfies:

$$\delta\ddot{\mathbf{r}} = \mathbf{a}_p + \frac{\mu}{\rho^3} \left[(1 - 2q)^{-3/2} \mathbf{r} - \delta\mathbf{r} \right] \quad (52)$$

where $q = \delta\mathbf{r} \cdot (2\mathbf{r} - \delta\mathbf{r})/r^2$.

Advantages:

- Faster than Cowell’s method (deviations change slowly)
- Better accuracy for small perturbations
- Less accumulation of roundoff error

Disadvantages:

- More complex formulation
- Requires *rectification* (resetting the osculating orbit) when $\delta r/r$ exceeds a tolerance
- The term $(1 - 2q)^{-3/2}$ requires careful numerical handling (difference of nearly equal quantities)

12.3 Störmer/Numerov vs. Cowell for Orbital Mechanics

Our `StörmerStep` and `NumerovStep` implementations are **integration methods**, while Cowell’s and Encke’s are **formulation frameworks**. The relationship:

Formulation	Integration Method	Use Case
Cowell’s	Störmer (order 2) or Numerov (order 4)	Direct integration of $y'' = f(x, y)$
Cowell’s	RK4, Adams-Moulton	General first-order reduction
Encke’s	Störmer/Numerov	Deviation from osculating orbit

Why Numerov outperforms standard Cowell integration:

1. **Order 4 accuracy** vs. order 2 for explicit Störmer or standard RK4 on first-order form
2. **Same stencil** (3 points) as explicit Störmer, but implicit correction achieves higher order
3. **Symplectic-like** energy behavior for near-Keplerian orbits (nearly circular)
4. **Fewer function evaluations** than equivalent RK4: 3 f -evals/step vs. 4 for RK4

Bate et al. [3] note that for orbital problems, the **Gauss-Jackson** (sum-squared, Σ^2) method is "one of the best, and most used, numerical integration methods for trajectory problems of the Cowell and Encke type." The Gauss-Jackson method is also a multistep predictor-corrector for second-order equations, similar in spirit to Numerov but with higher order and specialized for orbital mechanics.

12.4 Common Perturbations in Astrodynamics

Based on Curtis [4], Hintz [5], and Bate [3], the primary perturbations affecting Earth satellites are:

1. **Earth Oblateness (J_2):** The Earth's equatorial bulge causes secular variations in Ω (RAAN) and ω (argument of perigee). The nodal regression rate is:

$$\dot{\Omega} = -\frac{3J_2}{2} \sqrt{\frac{\mu}{a^3}} \left(\frac{R_\oplus}{a(1-e^2)} \right)^2 \cos i \quad (53)$$

2. **Atmospheric Drag:** Significant for LEO satellites. The acceleration is:

$$\mathbf{a}_d = -\frac{1}{2} \rho v_{rel}^2 \frac{C_D A}{m} \frac{\mathbf{v}_{rel}}{|\mathbf{v}_{rel}|} \quad (54)$$

where ρ is atmospheric density, $C_D \approx 2.0$ – 2.2 for typical satellites.

3. **Third-Body Gravity (Moon, Sun):** The lunar perturbation acceleration on an Earth satellite is:

$$\mathbf{a}_p = \mu_M \left(\frac{\mathbf{r}_{ms}}{r_{ms}^3} - \frac{\mathbf{r}_{Me}}{r_{Me}^3} \right) \quad (55)$$

where $\mathbf{r}_{ms}$ is the vector from Moon to satellite, $\mathbf{r}_{Me}$ from Moon to Earth.

4. **Solar Radiation Pressure:** For spacecraft with large area-to-mass ratio:

$$\mathbf{a}_{SRP} = -p_{SR} \frac{A}{m} \cos \theta \hat{\mathbf{s}} \quad (56)$$

where $p_{SR} \approx 4.5 \times 10^{-6}$ N/m² at 1 AU.

Special vs. General Perturbations:

- **Special perturbations:** Direct numerical integration (Cowell, Encke) — computes specific trajectories for given initial conditions.
- **General perturbations:** Analytic integration of series expansions (variation of parameters, Lagrange planetary equations) — provides analytical insight but is more complex.

12.5 Application: Numerov Orbit Propagation Demo

The `NumerovOrbitDemo.cpp` implementation demonstrates two scenarios:

Toy Example: Pure Two-Body LEO (400 km, 28.5° inclination)

- **Purpose:** Verify Numerov implementation correctness
- **Physics:** Central gravity only ($\mathbf{a} = -\mu\mathbf{r}/r^3$)
- **Validation:** Energy conservation to $\sim 0.04\%$ over 10 orbits
- **Expected:** Perfectly periodic orbit with constant energy

Real-World Example: LEO with J_2 Perturbation

- **Purpose:** Demonstrate dominant LEO perturbation effect
- **Physics:** Two-body + Earth oblateness ($J_2 = 1.08263 \times 10^{-3}$)
- **Perturbation formula** (Curtis §10.5):

$$\mathbf{a}_{J_2} = -\frac{3\mu J_2 R_\oplus^2}{2r^5} \begin{bmatrix} x(1 - 5z^2/r^2) \\ y(1 - 5z^2/r^2) \\ z(3 - 5z^2/r^2) \end{bmatrix} \quad (57)$$

- **Observable effects:**
 - Nodal regression: $-7.08/\text{day}$ (Curtis Eq. 4.52)
 - Perigee advance: $+11.52/\text{day}$ (Curtis Eq. 4.53)
- **Why nontrivial:** J_2 is the dominant LEO perturbation ($\sim 90\%$ of all perturbation effects)

Build and run (from Cosmos/Source):

```
mkdir -p build && cd build
cmake .. && make NumerovOrbitDemo
./Examples/NumerovOrbitDemo
```

Output: `numerov_twobody.csv`, `numerov_j2.csv` with columns `t,orbit,r_x,r_y,r_z,energy,error`.

13 Summary

1. Störmer’s rule exploits $y'' = f(x, y)$ (no y' in f) via the three-point recurrence $y_{m+1} = 2y_m - y_{m-1} + h^2 f_m$.
2. In δ -form: $\delta_m = \delta_{m-1} + h^2 f_m$, with starting value $\delta_0 = h\nu_0 + \frac{h^2}{2}f_0$. This form is mandatory to avoid numerical instability from the double root $\varrho(\zeta) = (\zeta - 1)^2$ (Section 3.3).
3. The algorithm is equivalent to Störmer-Verlet (leapfrog), which is symplectic for Hamiltonian systems.
4. “Order p ” means global error $\mathcal{O}(h^p)$ in step size h (not polynomial degree). The explicit method is order 2; Numerov’s implicit method (40) saturates the order barrier for the two-step stencil at order 4.
5. Gragg’s theorem guarantees an even-power error series in h , enabling efficient Bulirsch-Stoer Richardson extrapolation.
6. NR3’s `StepperStoerm` implements the $k = 2$ explicit Störmer in one-step extrapolation mode (HNW §II.14, Eq. 14.32) — the simplest special case of the general multistep family (HNW §III.10).
7. Numerov PECECE (Section 11) achieves order 4 on the same three-point stencil at 3 f -evaluations/step vs. 1 for explicit Störmer. Implemented as `NumerovStep`.

References.

References

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14 Gravitational Potential and Spherical Harmonics Expansion

This section derives the full spherical harmonic expansion of Earth’s gravitational potential from first principles, obtains the explicit J_2 and J_3 perturbation terms, and provides the Cartesian acceleration expressions used in orbit propagators such as Cosmos GNC.

14.1 Coordinate Conventions

We adopt the standard right-handed *spherical coordinates* (r, θ, ϕ) , where

- $r \geq 0$: radial distance from the Earth’s centre of mass,
- $\theta \in [0, \pi]$: *colatitude* (polar angle measured from the north pole along the rotation axis $+z$),
- $\phi \in [0, 2\pi)$: *east longitude*.

These relate to Cartesian coordinates (x, y, z) — with z pointing north along Earth's spin axis — by

$$x = r \sin \theta \cos \phi, \quad (58)$$

$$y = r \sin \theta \sin \phi, \quad (59)$$

$$z = r \cos \theta. \quad (60)$$

Inverting:

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \arccos \frac{z}{r}, \quad \phi = \text{atan2}(y, x). \quad (61)$$

Remark 5 (Latitude vs. colatitude). *Geodetic latitude φ satisfies $\varphi = \frac{\pi}{2} - \theta$, so $\cos \theta = \sin \varphi$ and $\sin \theta = \cos \varphi$. Some references (Vallado [6], Kaula [7]) use latitude; we use colatitude throughout for cleaner Legendre notation.*

14.2 Transformation Matrix: Spherical to Cartesian

The spherical orthonormal basis at (r, θ, ϕ) is

$$\hat{e}_r = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}, \quad (62)$$

$$\hat{e}_\theta = \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z}, \quad (63)$$

$$\hat{e}_\phi = -\sin \phi \hat{x} + \cos \phi \hat{y}. \quad (64)$$

These three directions are mutually orthogonal and right-handed: $\hat{e}_r \times \hat{e}_\theta = \hat{e}_\phi$. The matrix whose *columns* are the spherical basis vectors expressed in Cartesian components is

$$R(\theta, \phi) = \begin{pmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{pmatrix}, \quad (65)$$

so that a vector $\mathbf{v}$ with spherical components $(v_r, v_\theta, v_\phi)^T$ has Cartesian components

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = R(\theta, \phi) \begin{pmatrix} v_r \\ v_\theta \\ v_\phi \end{pmatrix}. \quad (66)$$

Since R is orthogonal ($R^T = R^{-1}$, verified by direct multiplication $R^T R = I_3$), the inverse is

$$\begin{pmatrix} v_r \\ v_\theta \\ v_\phi \end{pmatrix} = R^T(\theta, \phi) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}. \quad (67)$$

Remark 6 (Rotating frame). *The spherical frame $(\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi)$ rotates with the satellite's position: as θ and ϕ change along the orbit, R changes. When converting the spherical-frame acceleration components back to inertial Cartesian, apply (66) evaluated at the current (r, θ, ϕ) .*

14.3 Gravitational Potential: Multipole Expansion from First Principles

The Newtonian gravitational potential at exterior point $\mathbf{r}$ due to a body with mass density $\rho(\mathbf{r}')$ is

$$V(\mathbf{r}) = -G \int_{\mathcal{B}} \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV'. \quad (68)$$

(Convention: $V < 0$ for bound orbits. Some references use $U = -V > 0$.)

Legendre generating function. For $r = |\mathbf{r}| > r' = |\mathbf{r}'|$ (exterior expansion), the identity

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r} \right)^n P_n(\cos \gamma), \quad (69)$$

where γ is the angle between $\mathbf{r}$ and $\mathbf{r}'$, follows from the generating function of Legendre polynomials $\sum_{n=0}^{\infty} t^n P_n(u) = (1 - 2tu + t^2)^{-1/2}$ with $t = r'/r$, $u = \cos \gamma$. The series converges absolutely for $r > r'$.

Substituting (69) into (68):

$$V(\mathbf{r}) = -\frac{G}{r} \sum_{n=0}^{\infty} \frac{1}{r^n} \int_{\mathcal{B}} \rho(\mathbf{r}') (r')^n P_n(\cos \gamma) dV'. \quad (70)$$

Spherical harmonic addition theorem. The addition theorem for P_n decouples the angles of $\mathbf{r}$ and $\mathbf{r}'$:

$$P_n(\cos \gamma) = \frac{4\pi}{2n+1} \sum_{m=-n}^n Y_{nm}^*(\theta', \phi') Y_{nm}(\theta, \phi), \quad (71)$$

where Y_{nm} are the complex spherical harmonics. Substituting into (70) and integrating over $\mathbf{r}'$, the *multipole moments* of the body appear naturally:

$$V(\mathbf{r}) = -G \sum_{n=0}^{\infty} \sum_{m=-n}^n \frac{4\pi}{(2n+1)r^{n+1}} \left[\int_{\mathcal{B}} \rho(\mathbf{r}') (r')^n Y_{nm}^*(\theta', \phi') dV' \right] Y_{nm}(\theta, \phi). \quad (72)$$

Real harmonic form. Converting to real tesseral harmonics via $Y_{nm}^c = \sqrt{2} \operatorname{Re}[Y_{nm}]$ and $Y_{nm}^s = \sqrt{2} \operatorname{Im}[Y_{nm}]$, and introducing the reference equatorial radius R_e and total mass M to non-dimensionalise the coefficients, one obtains the standard *geopotential expansion*:

$$\boxed{V(r, \theta, \phi) = \frac{\mu}{r} \sum_{n=0}^{\infty} \sum_{m=0}^n \left(\frac{R_e}{r} \right)^n P_{nm}(\cos \theta) \left[C_{nm} \cos(m\phi) + S_{nm} \sin(m\phi) \right]} \quad (73)$$

where $\mu = GM$ is the standard gravitational parameter, R_e is the mean equatorial radius, P_{nm} are the *associated Legendre functions* (see Section 14.4), and C_{nm} , S_{nm} are the dimensionless Stokes coefficients (gravity field harmonics).

Remark 7 (Sign convention). *Some references (Kaula [7]; older geodesy) define V with a positive sign (force potential $U = -V$). The convention in (73) gives $V < 0$ and acceleration $\mathbf{a} = -\nabla V$.*

14.4 Legendre Polynomials and Associated Legendre Functions

The *Legendre polynomials* $P_n(u)$ are the $m = 0$ solutions of Legendre's differential equation $(1 - u^2)P_n'' - 2uP_n' + n(n+1)P_n = 0$. The first few are:

$$P_0(u) = 1, \quad (74)$$

$$P_1(u) = u, \quad (75)$$

$$P_2(u) = \frac{1}{2}(3u^2 - 1), \quad (76)$$

$$P_3(u) = \frac{1}{2}(5u^3 - 3u), \quad (77)$$

$$P_4(u) = \frac{1}{8}(35u^4 - 30u^2 + 3). \quad (78)$$

They satisfy the orthogonality relation $\int_{-1}^1 P_n(u) P_{n'}(u) du = \frac{2}{2n+1} \delta_{nn'}$.

The *associated Legendre functions* (un-normalized, matching (73)) are defined by

$$P_{nm}(u) = (1 - u^2)^{m/2} \frac{d^m}{du^m} P_n(u), \quad n \geq 0, 0 \leq m \leq n. \quad (79)$$

Explicitly for the terms needed through $n = 3$:

$$P_{20}(u) = P_2(u) = \frac{1}{2}(3u^2 - 1), \quad (80)$$

$$P_{21}(u) = 3u\sqrt{1 - u^2}, \quad (81)$$

$$P_{22}(u) = 3(1 - u^2), \quad (82)$$

$$P_{30}(u) = P_3(u) = \frac{1}{2}(5u^3 - 3u), \quad (83)$$

$$P_{31}(u) = \frac{3}{2}(5u^2 - 1)\sqrt{1 - u^2}, \quad (84)$$

$$P_{32}(u) = 15u(1 - u^2), \quad (85)$$

$$P_{33}(u) = 15(1 - u^2)^{3/2}. \quad (86)$$

Harmonic taxonomy. The (n, m) harmonics are classified as:

- **Zonal** ($m = 0$): depend only on latitude; axisymmetric rings. Coefficient $J_n \equiv -C_{n0}$. Earth's oblateness is dominated by $J_2 \approx 1.08 \times 10^{-3}$.
- **Tesseral** ($0 < m < n$): depend on both latitude and longitude; “checkerboard” patterns on the sphere.
- **Sectoral** ($m = n$): depend primarily on longitude; “orange-peel” slices.

14.5 Explicit Expansion: Monopole through J_3

We expand (73) term by term, substituting $u = \cos \theta$ and the Cartesian relations (58)–(60).

14.5.1 $n = 0$: Monopole (Point-Mass)

$$V_0 = \frac{\mu}{r}, \quad (87)$$

with $C_{00} = 1$ by normalization (total-mass convention). This is the two-body Keplerian potential.

14.5.2 $n = 1$: Dipole (Vanishes at Centre of Mass)

The $n = 1$ terms, using $P_{10}(u) = u$ and $P_{11}(u) = \sqrt{1 - u^2}$, are:

$$V_1 = \frac{\mu R_e}{r^3} [C_{10} z + C_{11} x + S_{11} y]. \quad (88)$$

The coefficients C_{10} , C_{11} , S_{11} are proportional to the components of the body's centre of mass expressed in the body-fixed frame. *When the origin is chosen at the body's centre of mass*, $C_{10} = C_{11} = S_{11} = 0$, and $V_1 \equiv 0$. This is the standard assumption for geocentric orbit mechanics.

14.5.3 $n = 2$, $m = 0$: The J_2 Zonal Term

$$V_{20} = \frac{\mu}{r} \left(\frac{R_e}{r} \right)^2 P_2(\cos \theta) C_{20}. \quad (89)$$

Using (76) with $u = \cos \theta = z/r$:

$$P_2(\cos \theta) = \frac{1}{2}(3 \cos^2 \theta - 1) = \frac{1}{2} \left(\frac{3z^2}{r^2} - 1 \right) = \frac{2z^2 - x^2 - y^2}{2r^2}. \quad (90)$$

Defining $J_2 = -C_{20} > 0$ (Earth is oblate, so $C_{20} < 0$):

$$\boxed{V_{J_2} = -\frac{\mu J_2 R_e^2}{2r^3} (3 \cos^2 \theta - 1) = -\frac{\mu J_2 R_e^2}{2r^5} (2z^2 - x^2 - y^2)}. \quad (91)$$

The $n = 2$ tesseral ($m = 1$) and sectoral ($m = 2$) harmonics are:

$$V_{21} = \frac{3\mu R_e^2}{r^5} (C_{21} xz + S_{21} yz), \quad (92)$$

$$V_{22} = \frac{3\mu R_e^2}{r^5} [C_{22}(x^2 - y^2) + 2S_{22}xy]. \quad (93)$$

For Earth, $|C_{21}|, |S_{21}| < 10^{-9}$ (negligible); $C_{22} \approx 1.574 \times 10^{-6}$ encodes the equatorial ellipticity.

14.5.4 $n = 3$, $m = 0$: The J_3 Zonal Term

$$V_{30} = \frac{\mu}{r} \left(\frac{R_e}{r} \right)^3 P_3(\cos \theta) C_{30}, \quad J_3 = -C_{30}. \quad (94)$$

Using (77) with $u = \cos \theta = z/r$:

$$P_3(\cos \theta) = \frac{1}{2}(5 \cos^3 \theta - 3 \cos \theta) = \frac{z(5z^2 - 3r^2)}{2r^3}. \quad (95)$$

Therefore:

$$\boxed{V_{J_3} = -\frac{\mu J_3 R_e^3}{2r^7} z(5z^2 - 3r^2)}. \quad (96)$$

For Earth, $J_3 \approx -2.532 \times 10^{-6}$ (negative: slight southern-hemisphere mass excess, the “pear-shape” perturbation).

Numerical constants (WGS84 / EGM96).

$$\begin{aligned}\mu &= 3.986\,004\,418 \times 10^{14} \text{ m}^3 \text{ s}^{-2}, \\ R_e &= 6\,378\,137 \text{ m}, \\ J_2 &= 1.082\,63 \times 10^{-3}, \\ J_3 &\approx -2.532 \times 10^{-6}.\end{aligned}$$

The full zonal potential through J_3 in compact spherical form is:

$$V \approx \frac{\mu}{r} \left[1 - J_2 \left(\frac{R_e}{r} \right)^2 \frac{3 \cos^2 \theta - 1}{2} - J_3 \left(\frac{R_e}{r} \right)^3 \frac{5 \cos^3 \theta - 3 \cos \theta}{2} \right]. \quad (97)$$

14.6 Gradient and Acceleration

The gravitational acceleration is $\mathbf{a} = -\nabla V$. The gradient in spherical coordinates is

$$\nabla V = \frac{\partial V}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{e}_\phi. \quad (98)$$

14.6.1 Two-Body (Keplerian) Acceleration

$$\mathbf{a}_0 = -\frac{\mu}{r^2} \hat{e}_r = -\frac{\mu}{r^3} \mathbf{r}, \quad (99)$$

so in Cartesian: $\mathbf{a}_0 = -(\mu/r^3)(x, y, z)^T$.

14.6.2 J_2 Acceleration in Spherical Coordinates

Differentiating (91) with respect to r and θ :

$$\frac{\partial V_{J_2}}{\partial r} = \frac{3\mu J_2 R_e^2}{2r^4} (3 \cos^2 \theta - 1), \quad (100)$$

$$\frac{1}{r} \frac{\partial V_{J_2}}{\partial \theta} = \frac{3\mu J_2 R_e^2}{r^4} \cos \theta \sin \theta. \quad (101)$$

Since V_{J_2} is independent of ϕ , the $\hat{e}_\phi$ component is zero. The spherical-frame J_2 acceleration is:

$$a_{J_2,r} = -\frac{3\mu J_2 R_e^2}{2r^4} (3 \cos^2 \theta - 1), \quad (102)$$

$$a_{J_2,\theta} = -\frac{3\mu J_2 R_e^2}{r^4} \cos \theta \sin \theta, \quad (103)$$

$$a_{J_2,\phi} = 0. \quad (104)$$

14.6.3 J_2 Acceleration in Cartesian Coordinates

Apply the rotation (66):

$$\begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix}_{J_2} = R(\theta, \phi) \begin{pmatrix} a_{J_2,r} \\ a_{J_2,\theta} \\ 0 \end{pmatrix}. \quad (105)$$

Expanding with $\cos \theta = z/r$, $\sin \theta \cos \phi = x/r$, $\sin \theta \sin \phi = y/r$:

$$a_x^{J_2} = a_{J_2,r} \sin \theta \cos \phi + a_{J_2,\theta} \cos \theta \cos \phi = \cos \phi (a_{J_2,r} \sin \theta + a_{J_2,\theta} \cos \theta). \quad (106)$$

Substituting (102)–(103) and simplifying, the result is the compact form used in orbital mechanics codes:

$$\mathbf{a}_{J_2} = -\frac{3\mu J_2 R_e^2}{2r^5} \begin{pmatrix} x(1 - 5z^2/r^2) \\ y(1 - 5z^2/r^2) \\ z(3 - 5z^2/r^2) \end{pmatrix}. \quad (107)$$

Remark 8 (Equatorial check). *Setting $z = 0$ (equatorial orbit): $\mathbf{a}_{J_2} = -(3\mu J_2 R_e^2/2r^5)(x, y, 0)^T$. This is a radially outward perturbation in the equatorial plane, consistent with the oblate equatorial bulge attracting a test mass less strongly than a sphere of the same mass.*

Remark 9 (Polar check). *Setting $x = y = 0$, $z = r$ (polar pass): $\mathbf{a}_{J_2} = -(3\mu J_2 R_e^2/2r^5) \cdot z(3 - 5)\hat{z} = +(\mu J_2 R_e^2/r^5)z\hat{z}$. The J_2 acceleration is directed away from Earth at the poles, as expected (less equatorial mass overhead means slightly less gravitational pull).*

14.6.4 J_3 Acceleration in Cartesian Coordinates

Differentiating (96) and applying the same rotation procedure:

$$\mathbf{a}_{J_3} = -\frac{5\mu J_3 R_e^3}{2r^7} \begin{pmatrix} xz(3 - 7z^2/r^2) \\ yz(3 - 7z^2/r^2) \\ \frac{3r^2}{5} - 6z^2 + \frac{7z^4}{r^2} \end{pmatrix}. \quad (108)$$

Note that $a_x^{J_3}$ and $a_y^{J_3}$ are both proportional to z : the J_3 perturbation has no in-plane component at the equator ($z = 0$), only north–south asymmetry.

14.7 Summary: Total Acceleration Through J_3

The total gravitational acceleration in inertial Cartesian (ECI) coordinates, including the two-body term and zonal harmonics through J_3 , is:

$$\mathbf{a}(\mathbf{r}) = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_{J_2} + \mathbf{a}_{J_3} + \mathcal{O}(J_4), \quad (109)$$

with $\mathbf{a}_{J_2}$ from (107) and $\mathbf{a}_{J_3}$ from (108). For a low-Earth orbit satellite, $|\mathbf{a}_{J_2}|/|\mathbf{a}_0| \approx J_2(R_e/r)^2 \approx 10^{-3}$ while $|\mathbf{a}_{J_3}|/|\mathbf{a}_0| \approx 10^{-6}$.

This acceleration is the right-hand side of the second-order ODE (1) propagated by the Störmer/Numerov integrators described in the preceding sections.